Photogrammetry & LiDAR point cloud

What this tool produces · *A highly accurate, millimeter-precision 3D digital representation (point cloud) of the physical asset, revealing structural deformations and material decay invisible to the naked eye,.*

G1 · Proof of Impact

Dimensions

Heritage **Spatial**

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Preparing the workshop · *Photogrammetry & LiDAR point cloud*

G1 · Proof of Impact

Why this tool matters

Freezing a fragile moment in time with beams of light; digitally immortalizing the exact scars, textures, and geometry of ancient stones before they fade away.

Evidence we produce

A highly accurate, millimeter-precision 3D digital representation (point cloud) of the physical asset, revealing structural deformations and material decay invisible to the naked eye,.

Duration & Material

Duration · 1 to 5 days of scanning, 1 to 2 weeks of digital processing

Material · Terrestrial Laser Scanners (TLS), UAVs (drones), point cloud processing software (e.g., Leica Cyclone, RealityCapture)

Reference case

During the documentation of the Worms Synagogue in Germany, a combination of Terrestrial Laser Scanning (TLS), mobile SLAM scanning, and UAV photogrammetry was used. This multi-layered scanning approach captured precise geometries of both subterranean elements and inaccessible roofscapes, providing critical data for the site's structural monitoring and conservation,.

Suggested agenda · Brief (10 min) › Method walk-through (15 min) › Animation canvas (60–90 min) › Harvest (20 min)

Capture · process · publish · reuse

Duration
1 to 5 days of scanning, 1 to 2 weeks of digital proces...

1 Capture

plan flight paths
control-points, lighting

Equipment
Resolution
Output
Owner

2 Process

align, mesh, texture
clean noise

Equipment
Resolution
Output
Owner

3 Publish

point cloud, mesh, drawings
delivery formats

Equipment
Resolution
Output
Owner

4 Reuse

BIM, AR, monitoring
stewardship-ready

Equipment
Resolution
Output
Owner

Quality gates

- **Geo-referenced**
- **Within tolerance**
- **Documented metadata**
- **Open formats**
- **Archive plan**

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Harvesting our *learnings*

After running the canvas, capture three layers of insight.

What inspires me

Stand-out signals, surprising stories, ideas to remember

How to apply it here

Concrete adaptations to our context, our team, our constraints

What we need to be autonomous

Skills, allies, resources, decisions still to secure

Each participant fills a copy. Then debrief together to surface patterns and next steps.